

1. The perceptron update rule is an online learning algorithm for binary classification that updates parameters only when a sample is misclassified. It updates parameters using $w \leftarrow w + \eta y_i x_i$ and $b \leftarrow b + \eta y_i$, rotating the decision hyperplane toward misclassified positive points and away from negative ones. Under the Perceptron Convergence Theorem, this process is guaranteed to halt in finite steps if given the dataset is linearly separable.

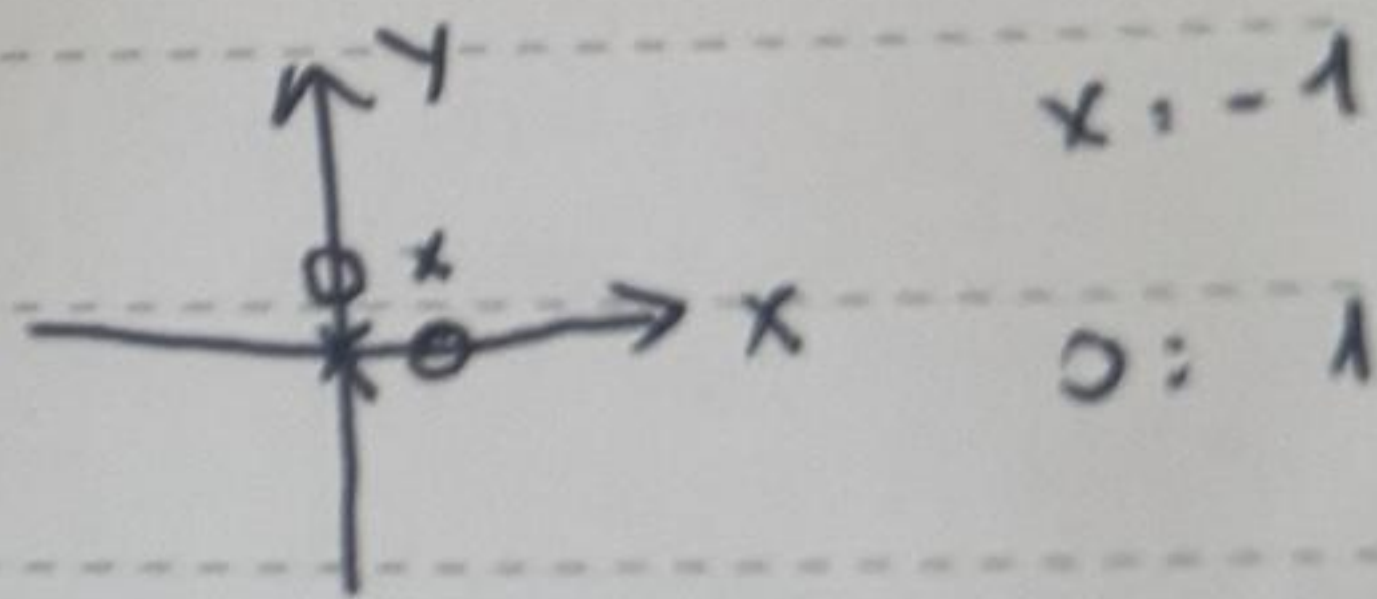
2. Linear separability is a geometric property of a binary dataset, where a single decision hyperplane $w^T x + b = 0$ can completely divide all positive samples from negative ones. In 2D, this means a straight line can separate 2 classes without misclassified points. Without it, Perceptron cannot converge.

3. the XOR function — $(0,0) \rightarrow -1$

$(0,1) \rightarrow 1$

$(1,0) \rightarrow 1$

$(1,1) \rightarrow -1$



$\Rightarrow$ not linearly separable

$\Rightarrow$ As stated in Q2, Perceptron fails this due to it ~~being~~ not being linearly separable.

Date: _____

4. Perceptron uses step function and updates weights on misclassifications.

→ fails to converge non-separable data

Delta rule uses gradient descent to minimize MSE, update weights on every sample based on error magnitude.

→ It converges to a minimum-error solution in any ~~data~~ datasets